

Creating a Basic IP PBX with Asterisk



Transforming the open source Asterisk software to a fully functional IP PBX is achieved with a few simple steps. In the previous sessions, we have familiarised ourselves with the general Asterisk environment as well as the common hardware involved. In this article, we will go into the details of creating a basic IP PBX.

We first need to decide on the configuration of the PBX —is it going to be a pure IP PBX or will it also connect to ISDN, GSM, etc? Let's select a very common configuration for the Indian scenario—an IP PBX for 100 subscribers with one PRI and four GSM trunks.

Let us start by defining the OS, which could theoretically be any Linux flavour like Debian, CentOS, Ubuntu, etc. We will base all further discussions on CentOS.

Next is the selection of the hardware. It's preferable to select server hardware rather than typical desktop hardware, as we expect the PBX to run 24 hours and have a lifespan of at least 5-10 years. In a 100-subscriber scenario, we can opt for an Intel dual core or a Core i3 with 4GB RAM and a 500GB hard disk drive. The 100-subscriber configuration can also work with 2GB RAM, but may have quality issues, when a lot of subscribers make calls simultaneously. Even the 4GB RAM set-up may have problems if all the 100 subscribers make calls at the same time. These values are based on my experience of the typical load in a 100-subscriber environment. For higher loads like 500 or 1000 subscribers, the use of higher RAM configurations and SSD hard disks for CDRs and log files is recommended.

After hardware selection and installation of the OS, we move on towards the software installation.

Asterisk is a software switching platform, capable of running on standardised PC hardware with the requisite accessories to connect to various telecom networks. In this third article in the series, the author explains how to create a basic IP private branch exchange (PBX).

Disabling SELinux

In CentOS, the security-enhanced Linux (SELinux) system is enabled by default, and it often gets in the way of Asterisk. There are multiple articles on the Internet that describe the correct configuration of SELinux, but we'll disable it for the sake of simplicity.

Installing basic libraries

The code given below will install OpenSSL security, the gcc compiler, the ncurses GUI library, the XML library, the SQLite database, etc.

```
yum install -y make wget openssl-devel ncurses-devel newt-
devel libxml2-devel kernel-devel gcc gcc-c++ sqlite-devel
libuuid-devel
```

These base libraries are required for subsequent modules.

Installing DAHDI

DAHDI (Digium/Asterisk Hardware Device Interface) is a framework for interfacing with digital telephony cards in Asterisk. It contains drivers for the interface and also tools for monitoring. As we are using a PRI card in our configuration, this installation needs to be done.

Download:

```
cd /usr/src/
wget http://downloads.asterisk.org/pub/telephony/DAHDI-linux-
complete/DAHDI-linux-complete-2.9.2+2.9.2.tar.gz
```

Unzip:

```
tar xzvf DAHDI-linux-complete-2.9.2+2.9.2.tar.gz
```

Install:

```
cd /usr/src/DAHDI-linux-complete-2.9.2+2.9.2
make
make install
make config
```

Installing libpri

libpri is required to handle the protocol part of the PRI interface.